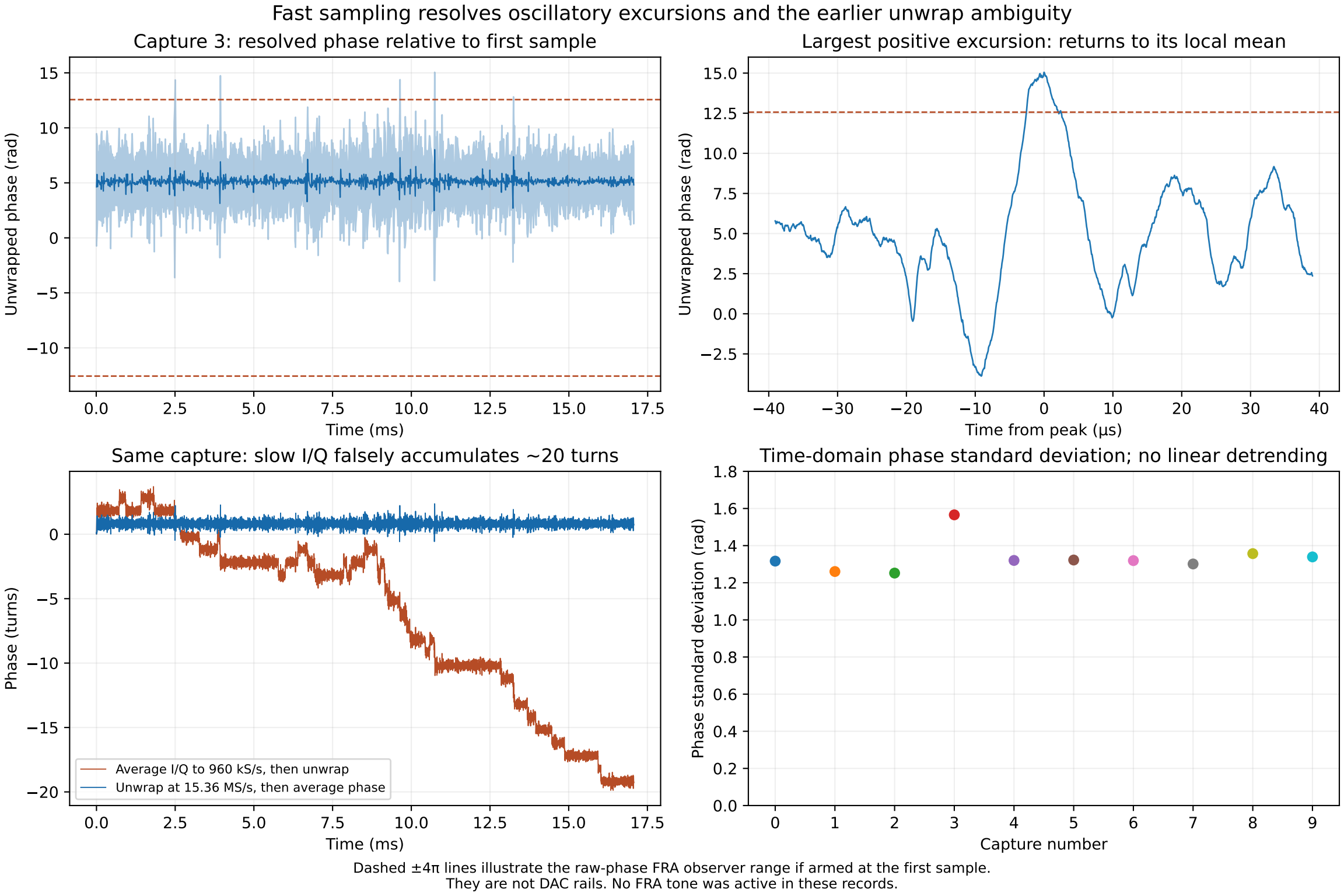

Fast sampling resolves oscillatory excursions and the earlier unwrap ambiguity
Capture 3: resolved phase relative to first sample
Unwrapped phase (rad)
Time (ms)
Largest positive excursion: returns to its local mean
Unwrapped phase (rad)
Time from peak (µs)
Same capture: slow I/Q falsely accumulates ~20 turns
Phase (turns)
Time (ms)
Average I/Q to 960 kS/s, then unwrap
Unwrap at 15.36 MS/s, then average phase
Time-domain phase standard deviation; no linear detrending
Phase standard deviation (rad)
Capture number
Dashed ±4π lines illustrate the raw-phase FRA observer range if armed at the first sample.
They are not DAC rails. No FRA tone was active in these records.